

CHAPTER 1: INTRODUCTION TO MACHINE LEARNING

Machine learning is a subfield of artificial intelligence focused on building systems that learn from data. Rather than explicitly programming every rule, machine learning algorithms iteratively adjust internal parameters to minimize loss on empirical observations.

A foundational taxonomy partitions machine learning into supervised learning, unsupervised learning, and reinforcement learning. In supervised learning, the model is provided with paired training instances (x, y) , where x denotes the feature vector and y denotes the target ground truth label.

SECTION 1.2: SUPERVISED OPTIMIZATION OBJECTIVES

Consider empirical risk minimization (ERM). Given a hypothesis class H and a convex surrogate loss function $L(f(x), y)$, the objective is to find parameters θ that minimize the regularized sample average loss:

$$\min_{\theta} (1/N) * \sum_{i=1}^N L(f_{\theta}(x_i), y_i) + \lambda R(\theta)$$

Here, λ represents the regularization hyperparameter controlling the trade-off between bias and variance.

SECTION 1.3: ARCHITECTURAL FLOW AND GRADIENT DESCENT

Below is a schematic diagram illustrating the forward inference pass, error calculation, and backward automatic differentiation pass via stochastic gradient descent (SGD).

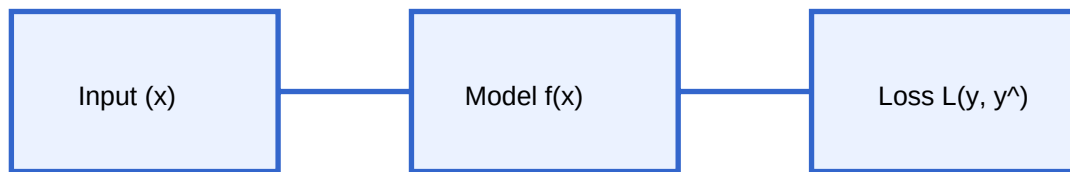


Figure 1.1: Canonical Supervised Learning Pipeline